



UNIVERSITY OF THE PUNJAB

Eight Semester 2015
Examination B.S. 4 Years Programme

Roll No.

PAPER: Statistical Inference-II (Theory)
Course Code: STAT-407

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided

SUBJECTIVE

- Q.No.2. Write short answer to each of the following questions. (20)
- Write the properties of moment estimators. 496 (4)
 - What modification(s) in minimum Chi-square method of estimation give rise in the method of modified minimum Chi-square? 434 (2)
 - Clarify the concept of shortest confidence interval. 472686814 (3)
 - What do you understand by confidence region? 509 (3)
 - State Neyman Pearson-lemma. 458 (3)
 - Distinguish between non-randomized and randomized test? (2)
 - What do you understand by Likelihood function, what is its utility? (7+3)

Q.No.3.

- a) Let $X_1, X_2, X_3, \dots, X_n$ be independent r.v's each with the lognormal distribution, find the lognormal distribution moment estimators of μ and σ^2 . 356

A ①

b) Assume that the daily sale of petrol follows exponential distribution. The hypothesis H_0 that the sale of petrol is 1000 liters per day is tested against the hypothesis that it is 1500 liters per day. If the sale on a day is 1200 liters or more, H_0 is rejected. Find the size of type I error, Type II error and power of the test? (10)

Q.No.4.

Let $X \sim N(0, \sigma^2)$ where σ^2 is known. If you are interested to test hypothesis $H_0: \mu = 100, H_1: \mu = 105$. Obtain the average sample number (ASN) for both state of nature. Assume $\alpha_n = 0.01, \beta_n = 0.05$ and $\sigma^2 = 100$ (5+)

Q.No.5.

- a) Based on random sample of size n from Bernoulli distribution obtain minimum χ^2 chi square estimator (MCSE) and MLE of 'p' also compare the two estimators. p 479
- b) What do you know about pivotal quantity and discuss its significance? p 565

②

Attempt this Paper on Separate Answer Sheet provided.

Part-2 (SUBJECTIVE)

Note: Attempt ALL questions.

- Q 2 Write the short answer to each of the followings
- ✓ Compare the moment estimator and MLE? (04)
 - ✓ Compare Maximum likelihood estimator and a BLUE estimator (03)
 - ✓ In which situation the Lloyd's method is used? S23 (02)
 - ✓ Write the situation in which minimum chi square method is used. -491 (03)
 - ✓ What is the difference between inference and Bayesian inference? -485 (02)
 - ✓ Write different types of Loss function (03)
 - ✓ What do you mean by Average Sample Number (ASN)? In which situation it is used? S10 (03)
- Q 3 Based on random sample of size n from Poisson distribution obtain moment estimator, minimum variance estimator (MCSE) and MLE of θ and compare the three estimators. (12)
- A
- Q 4 Let $x_1, x_2, \dots, x_n$ denote a random sample from Bernoulli distribution. Assume that the prior distribution of θ is given by $g(\theta) = I_{(0,1)}(\theta)$. Find the posterior Bayes estimators of θ and $\tau(\theta) = \theta(1-\theta)$. (06)
- P 411 ✓ ✓ ✓
- Q 5 Let $X_1, X_2, \dots, X_n$ be a random sample from $f(x; \theta) = \theta^x (1-\theta)^{1-x} I_{(0,1)}(x)$ where $\theta = \theta_0$ or $\theta = \theta_1$, $\theta_0 < \theta_1$. Apply the Neyman-Pearson Lemma to test $H_0: \theta = \theta_0$ versus $H_1: \theta = \theta_1$. (1)
- A
- P 015 ✓



$$\sum_{i=1}^n \sum_{j=1}^{k_i} \frac{(P_{ij} - \pi_{ij})^2}{\pi_{ij}}$$



UNIVERSITY OF THE PUNJAB

Eighth Semester - 2017
Examination: B.S. 4 Years Programme

Roll No.

PAPER: Statistical Inference-II (Theory)
Course Code: STAT-407

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

Q.2 Write short and brief answer to the followings.

- Why moment estimator is considered important even in this modern era of computerization? (05)
- Differentiate between confidence interval and large sample confidence interval. (04)
- What is the utility of likelihood function in MLE? (02)
- What do you mean by 95% confidence interval for a parameter? (03)
- Compare the properties of MLE and OLS estimate. (04)
- Why we have to use sequential sampling? (02)

Q.3. Let $P(X = x) = \binom{x + \theta_2 - 1}{x} \theta_1^{\theta_2} (1 - \theta_1)^x$ $x = 0, 1, \dots, \infty$ (10)

Find the moment estimators of θ_1 and θ_2

Q.4. Let $x_1, x_2, \dots, x_n$ denote a random sample from a distribution which has p.d.f. $f(x)$ that is positive on only non negative integers. It is desired to test the simple hypothesis $H_0: f(x) = e^{-1}/x!$, $x = 0, 1, 2, \dots$ against alternative simple hypothesis $H_1: f(x) = (1/2)^{x+1}$, $x = 0, 1, 2, \dots$. Derive the expression for BCR (Best critical region). Consider the case of $n=1$ and $k=1$, k being any positive integer in the expression $\frac{L(\theta', x_1, x_2, \dots, x_n)}{L(\theta'', x_1, x_2, \dots, x_n)} \leq k$ where $H_0: \theta = \theta'$, $H_1: \theta = \theta''$. (12)

Find the power of the test for this combination of n and k when
(i) H_0 is true (ii) H_1 is true

Q.5. Obtain the MLE for λ and λ^2 from the Rayleigh distribution with p.d.f $f(x) = (2x/\lambda^2) \exp[-x^2/\lambda^2]$, $0 < x < \infty$ so prove that MLE for this problem is Invariant. (08)





UNIVERSITY OF THE PUNJAB

Eighth Semester – 2018

Examination: B.S. 4 Years Program

Roll No. 010606

PAPER: Statistical Inference-II (Theory)

Course Code: STAT-407 Part – II

MAX. TIME: 2 Hrs. 45 Min.

MAX. MARKS: 50

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2. Write the short answer to each of the followings

- Readily availability of moment estimators is useful in which processes while consistency of these estimators is useful in which procedures. (02)
- Compare Maximum likelihood estimator and a ordinary least square estimator. (05)
- Differentiate between confidence belt and confidence region. (03)
- Differentiate between relational and non-relational hypothesis, in which type of situations these are used? (04)
- Write the situations when likelihood ratio test is mandatory. (02)
- What is the relationship between Loss function and Risk function? (02)
- Differentiate between most powerful test and the power of the test. (02)

Q.3. Let $x_1, x_2, \dots, x_n$ be a random sample from beta distribution of the form $f(x; \theta) = \theta(1+\theta)(1-x)x^{\theta-1}$ $0 < x < 1$, for some $\theta > 0$ estimate θ by method of moments. Also find the variance of θ . (10)

Q.4. Let $x_1, x_2, x_3, \dots, x_n$ be a random sample from the density $f(x; \theta) = 1/\theta, 0 < x < \theta$. Let $Y_n = \max[x_1, x_2, x_3, \dots, x_n]$ be sufficient statistic also it is MLE of θ for which $f_T(t, \theta) = n \left(\frac{t}{\theta} \right)^{n-1} \cdot \frac{1}{\theta}, 0 < t < \theta$. Find the confidence interval for θ using statistical method. (10)

Q.5. Let $x \sim N(\theta, \sigma^2)$ where σ^2 is known, if you are interested to test $H_0: \theta = \theta_0 = 100$ against $H_1: \theta = \theta_1 = 105$, obtain Average sample number (ASN) for both states of nature. Assume $\alpha_a = 0.01, \beta_a = 0.05, \sigma^2 = 100$ (10)





ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Question No.2. Answer the following.

(4 × 5 = 20)

- What is the rationale of moment estimation? Under what situation moment estimators and maximum Likelihood estimators are equally efficient?
- Write down the assumptions and properties of Least Squares Estimation method.
- Differentiate between Most Powerful and Uniformly Most Powerful Tests.
- Suppose $X_1, X_2, X_3, \dots, X_n$ constitutes a random sample from Bernoulli distribution with

$$f(x; \theta) = \theta^x (1 - \theta)^{1-x} ; x = 0, 1.$$

Obtain the MLE estimator of θ .

Question No. 3. Obtain the MLE of the unknown parameter θ for the following distribution, (12)

$$f(x; \theta) = \frac{1}{\Gamma_P \theta^P} x^{P-1} e^{-x/\theta} ; x > 0$$

Where P is known. Obtain the variance of the estimator and check it for Consistency, Efficiency and Sufficiency.

Question No. 4. State and Prove Neyman-Pearson Lemma.

(10)

Question No. 5. Define pivotal quantity. Consider a random sample of size "n" from a

(08)

distribution having p.d.f $f(x; \beta) = \beta e^{-\beta x} ; x > 0, \beta > 0$.

Find 95% Large sample confidence interval for β .





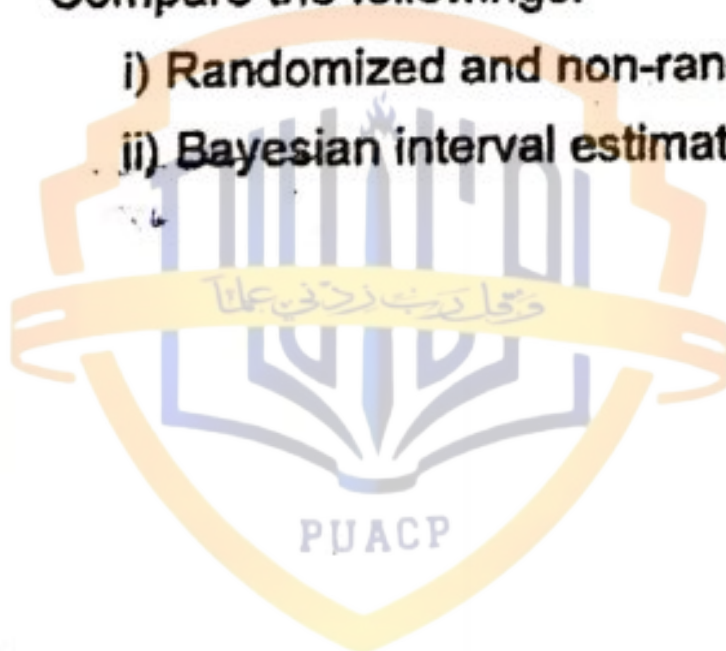
ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2. Answer the following short questions.

- i) What is the principle of maximum likelihood estimators? (02)
- ii) Compare Maximum likelihood estimator and an ordinary least square estimator. (05)
- iii) Differentiate between confidence belt and confidence region. (03)
- iv) Differentiate between relational and non-relational hypothesis, in which type of situations these are used? (04)
- v) Write the situations when likelihood ratio test is mandatory. (02)
- vi) What is the relationship between Loss function and Risk function? (02)
- vii) Differentiate between most powerful test and the power of the test. (02)

Answer the following questions.

- Q.3. Let $x_1, x_2, \dots, x_n$ be a random sample from beta distribution of the form $f(x; \theta) = \theta(1+\theta)(1-x)x^{\theta-1}$ $0 < x < 1$, for some $\theta > 0$ estimate θ by method of moments. Also find the variance of θ . (08)
- Q.4. Let $x \sim \text{Bernoulli}$ population, using sequential probability ratio test then test $H_0: \theta = 1/3$ vs $H_1: \theta = 2/3$. (07)
- Q.5. Let $x \sim N(\theta, \sigma^2)$ where σ^2 is known, if you are interested to test $H_0: \theta = \theta_0 = 100$ against $H_1: \theta = \theta_1 = 105$, obtain Average sample number (ASN) for both states of nature. Assume $\alpha_0 = 0.01$, $\beta_0 = 0.05$, $\sigma^2 = 100$ (09)
- Q.6. Compare the followings: (06)
 - i) Randomized and non-randomized tests.
 - ii) Bayesian interval estimates and large sample confidence intervals.





UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Eighth Semester – Spring 2022

Paper: Statistical Inference-II (Theory)

Course Code: STAT-407

Roll No. 026473

Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Write short answers to the following questions. (6x5=30)

- i) Write the properties of maximum likelihood estimators.
- ii) Compare Maximum likelihood estimator and an ordinary least square estimator.
- iii) Describe the procedure of calculation of the variance of the parameters by ordered least square estimator and least square estimator.
- iv) Differentiate between relational and non-relational hypothesis, in which type of situations these are used?
- v) Write short note on sequential probability ratio test.
- vi) Write short note on Average Sample Number.

Write detailed answers to the following questions. (3x10=30)

Q.2. Find the ML estimator of " θ " in the Cauchy distribution, also find the asymptotic variance of its estimator.

Q.3. Let " x " be a random sample $f(x) = \theta^x (1 - \theta)^{1-x}$; Where " θ " is distributed a priori as uniform over (0, 1). Find the Bayes Estimator of " θ " with respect to the loss function. $l(\theta, t) = (\theta - t)^2 / \theta^2$.

Q.4. Let $x_1, x_2, x_3, \dots, x_n$ denote a random sample from a distribution which has pdf $f(x)$ that is positive on only non-negative integers. It is desired to test the simple hypothesis. $H_0 = \frac{e^{-1}}{x!}; x = 0, 1, 2, 3, \dots$ against the alternative

$$H_1 = \left(\frac{1}{2}\right)^{x+1}; x = 0, 1, 2, 3, \dots$$

Derive the expression for BCR, consider the case of $n=1$ and $k=1$, Where " k " is any positive integer in the expression

$$\frac{L(\theta', x_1, x_2, x_3, \dots, x_n)}{L(\theta'', x_1, x_2, x_3, \dots, x_n)} \leq k$$

Where $H_0: \theta = \theta', H_1: \theta = \theta''$. Find the power of test for this combination of " n " and " k ".

- a) When H_0 is true
- b) When H_1 is true.

Q.1 A random variable X takes the values 0, 1 and 2 with respective probabilities (40)

$$\left[\frac{\theta}{4N} + \frac{1}{2} \left(1 - \frac{\theta}{N} \right) \right], \left[\frac{\theta}{2N} + \frac{\alpha}{2} \left(1 - \frac{\theta}{N} \right) \right] \text{ and } \left[\frac{\theta}{4N} + \frac{1-\alpha}{2} \left(1 - \frac{\theta}{N} \right) \right],$$

Where N is a known number and 'α' and 'θ' are unknowns.
If 75 independent observations on X yield the values 0, 1 and 2 with frequencies 27, 38 and 10 respectively. Estimate θ and α by method of moments.

Q.2 In testing $H_0: \mu=20$ VS $H_1: \mu > 20$ with $\alpha=0.05$ based on the sample of $n=30$. The sample mean and standard deviation are 23 and 5 respectively. (40)

- Test the hypothesis and find the confidence interval also.
- Sketch the power curve for $\mu=21, 22, 23, 24, 25, 26$.

Q.3 Viva (10)

Q.4 Note Book (10)

$$\begin{aligned} \frac{3\theta}{2N} - \frac{2\alpha}{N} &= \frac{\theta}{2N} + \frac{\alpha}{N} \\ \frac{3\theta - 4\alpha}{2N} &= \frac{\theta + \alpha}{2N} \\ -\frac{\alpha\theta}{2N} + \frac{2\alpha\theta}{N} &= \frac{\alpha - 2\alpha}{2} = -\frac{\alpha}{2} \\ \frac{-\alpha\theta + 4\alpha\theta}{2N} &= \frac{3\alpha\theta}{2N} \\ \frac{\alpha - 4\alpha}{2} &= -\frac{3\alpha}{2} \end{aligned}$$

$$\begin{aligned} \frac{\theta}{2N} + \frac{\alpha}{2N} &= \frac{\theta + \alpha}{2N} = \frac{11\theta}{2N} \\ \frac{\alpha}{2} - \alpha &= \frac{\alpha - 2\alpha}{2} = -\frac{\alpha}{2} \\ -\frac{\alpha\theta}{2N} + \frac{\alpha\theta}{N} &= \frac{-\alpha\theta + 2\alpha\theta}{2N} \\ &= \frac{\alpha\theta}{2N} \end{aligned}$$



THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Write short answers to the following questions.

(6x5=30)

- I. What is Average Sample Number (ASN). How it is computed when Null Hypothesis is true.
- II. Let $X_1, X_2, \dots, X_n$ denote the random sample from Poisson distribution with parameter λ . Find MLE of λ .
- III. Differentiate between Prior Distribution and Posterior Distribution.
- IV. What is Confidence Region.
- V. Compare Best Linear Unbiased Estimator (BLUE) and Minimum Variance Quadratic Unbiased Estimator (MVQUE).
- VI. Make Comparison of Minimum Chi-Square Estimator (MCSE) and Maximum Likelihood Estimator (MLE).

Answer the following questions.

(3x10=30)

Question 2: Estimate α and β using Moment Estimation from the Pearson Type-II distribution having the p.d.f

$$f(x; \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-x\beta}, \quad 0 < x < \infty$$

Question 3: Let $X \sim N(\theta, 100)$, perform SPRT to test

$$H_0: \theta = 75 \text{ v/s } H_1: \theta = 78.$$

Such that α and β are approximately equal to 0.10.

Question 4: Based upon the random sample of size n from

$$f(x; \theta) = \theta^x (1 - \theta)^{1-x}, \quad x = 0, 1$$

Obtain the MCSE for θ